

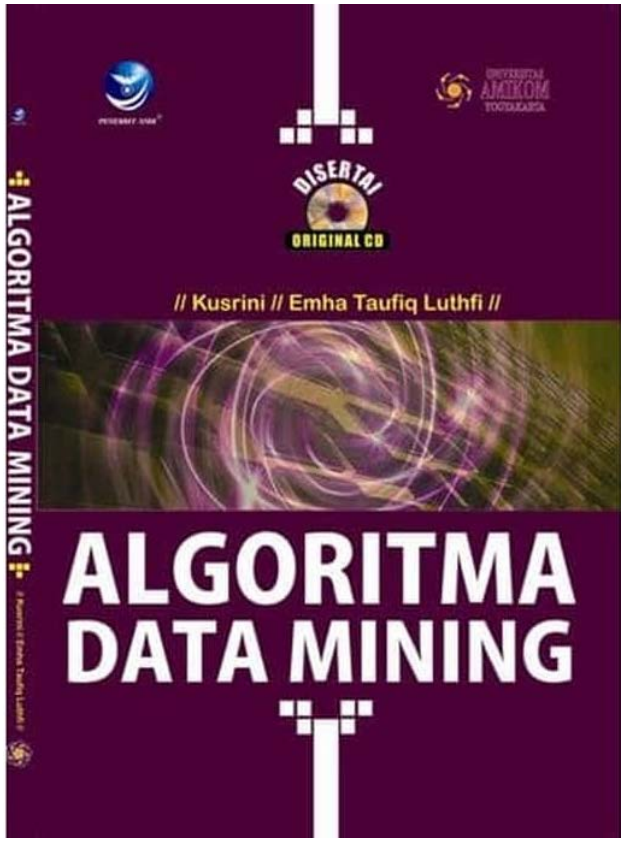
# Data Mining
# Klasifikasi dengan C.45

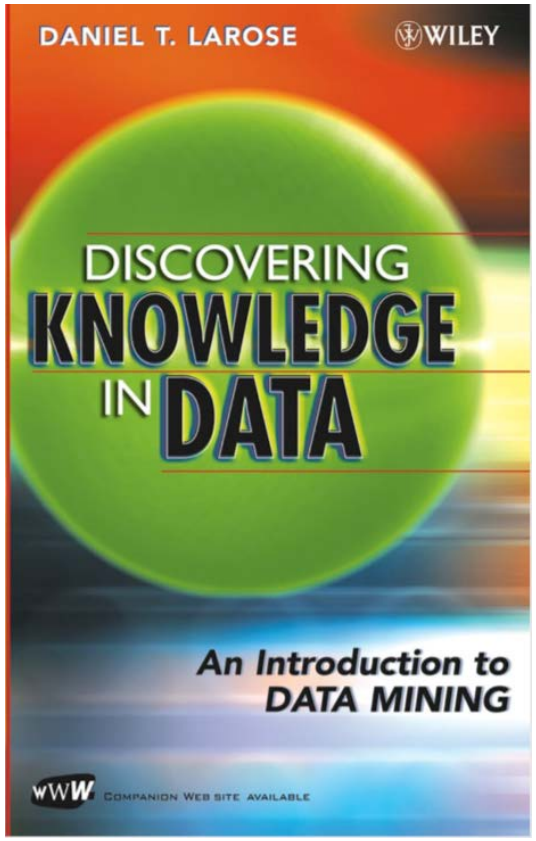
Dr. Kusrini, M.Kom

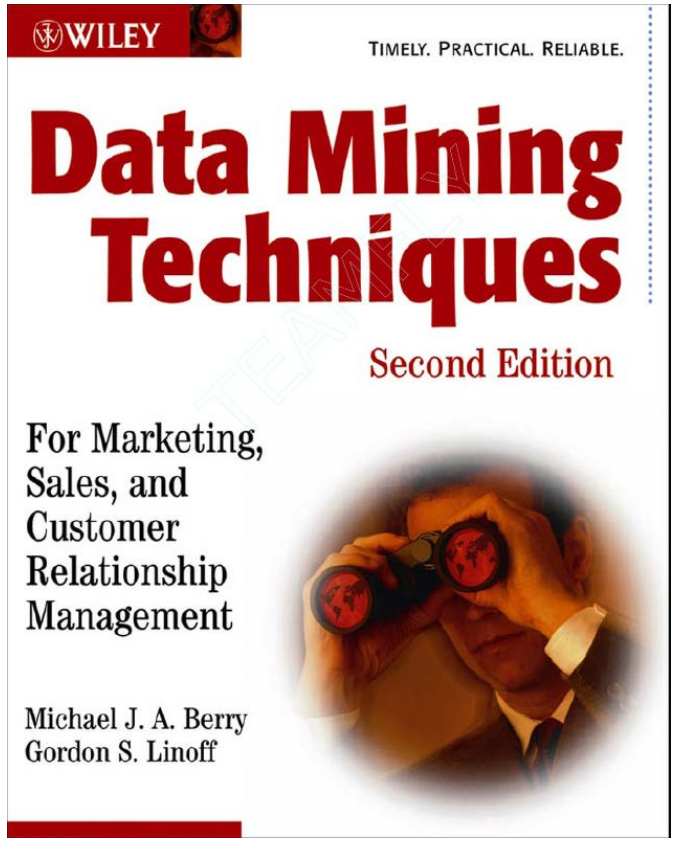
Universitas AMIKOM Yogyakarta

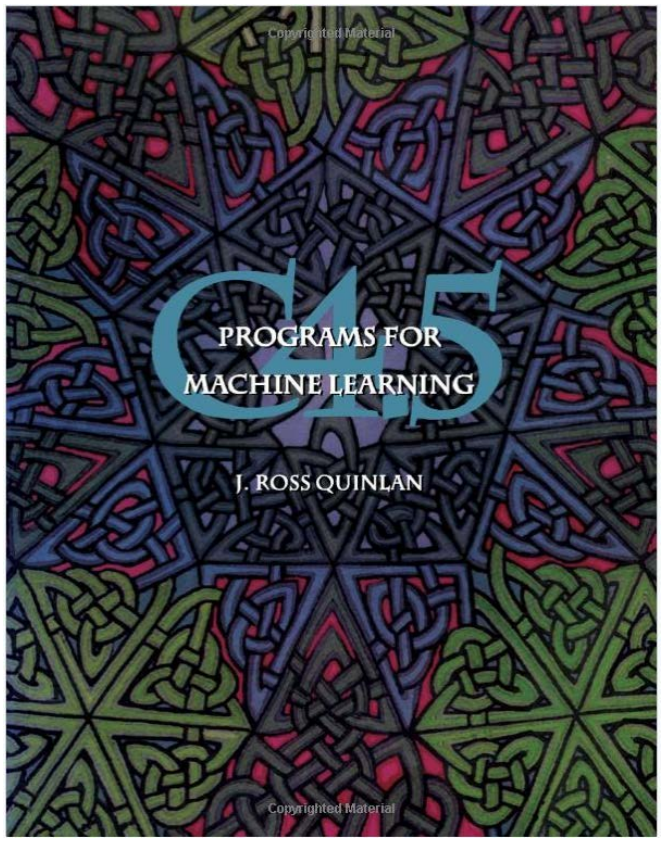
kusrini@amikom.ac.id

# References


ALGORITMA DATA MINING
// Kusrini // Emha Taufiq Luthfi //
DISERTAI
ORIGINAL CD
ALGORITMA DATA MINING

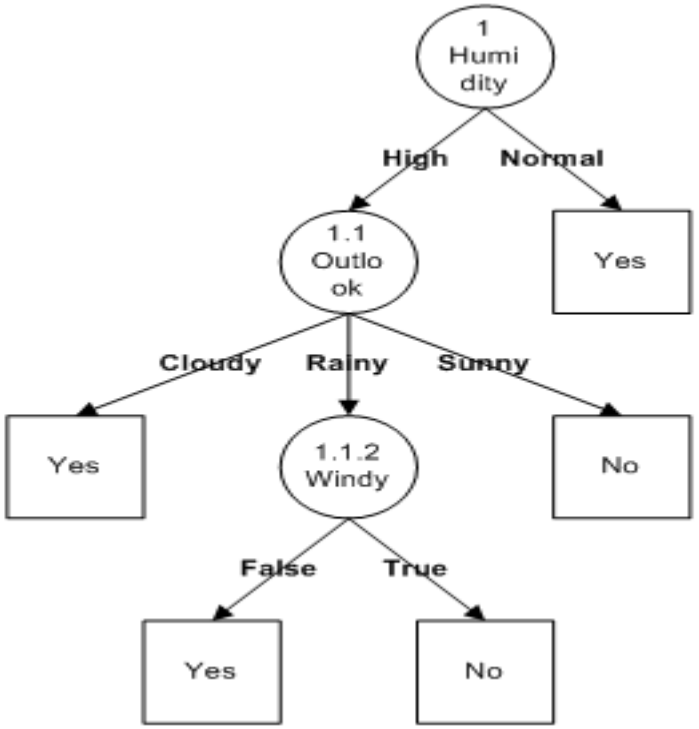

DANIEL T. LAROSE
WILEY
DISCOVERING
KNOWLEDGE
IN DATA
An Introduction to
DATA MINING
COMPANION WEB SITE AVAILABLE


WILEY
TIMELY. PRACTICAL. RELIABLE.
Data Mining
Techniques
Second Edition
For Marketing,
Sales, and
Customer
Relationship
Management
Michael J. A. Berry
Gordon S. Linoff


C4.5
PROGRAMS FOR
MACHINE LEARNING
J. ROSS QUINLAN

# Pohon Keputusan

- Pohon keputusan merupakan sebuah struktur yang dapat digunakan untuk membagi kumpulan data yang besar menjadi himpunan-himpunan record yang lebih kecil dengan menerapkan serangkaian aturan keputusan. Dengan masing-masing rangkaian pembagian, anggota himpunan hasil menjadi mirip satu dengan yang lain (Berry, Michael J.A., Linoff, Gordon S., 2004)
- Sebuah model pohon keputusan terdiri dari sekumpulan aturan untuk membagi sejumlah populasi yang heterogen menjadi lebih kecil, lebih homogen dengan memperhatikan pada variabel tujuannya



# Langkah pembuatan pohon keputusan

- Pilih atribut sebagai root
- Buat cabang untuk masing-masing nilai
- Bagi kasus dalam cabang
- Ulangi proses untuk masing-masing cabang sampai semua kasus pada cabang memiliki kelas yang sama

# Pemilihan root node

$$Gain(S, A) = Entropy(S) - \sum_{i=1}^{n} \frac{|S_i|}{|S|} * Entropy(Si) .$$

Dengan :
S : Himpunan kasus
A : Atribut
n : Jumlah partisi atribut A
|Si| : Jumlah kasus pada partisi ke i
|S| : Jumlah kasus dalam S

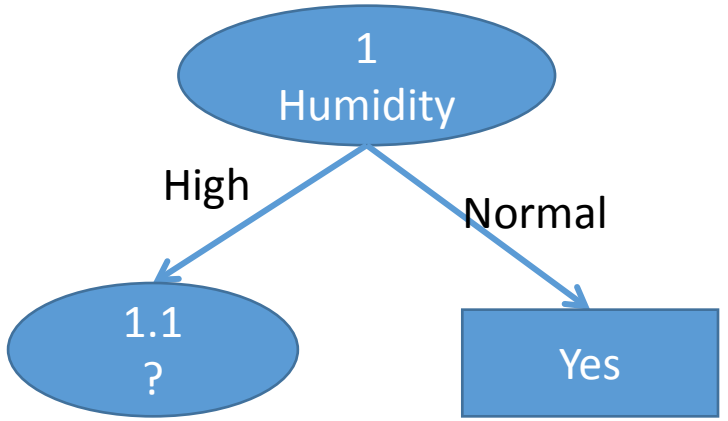
$$Entropy(S) = \sum_{i=1}^{n} - pi * \log_2 pi \ \ldots\ldots$$

dengan :
S : Himpunan Kasus
A : Fitur
n : Jumlah partisi S
$p_i$ : Proporsi dari $S_i$ terhadap S

| NO | OUTLOOK | TEMPERATURE | HUMIDITY | WINDY | PLAY |
|---|---|---|---|---|---|
| 1 | Sunny | Hot | High | FALSE | No |
| 2 | Sunny | Hot | High | TRUE | No |
| 3 | Cloudy | Hot | High | FALSE | Yes |
| 4 | Rainy | Mild | High | FALSE | Yes |
| 5 | Rainy | Cool | Normal | FALSE | Yes |
| 6 | Rainy | Cool | Normal | TRUE | Yes |
| 7 | Cloudy | Cool | Normal | TRUE | Yes |
| 8 | Sunny | Mild | High | FALSE | No |
| 9 | Sunny | Cool | Normal | FALSE | Yes |
| 10 | Rainy | Mild | Normal | FALSE | Yes |
| 11 | Sunny | Mild | Normal | TRUE | Yes |
| 12 | Cloudy | Mild | High | TRUE | Yes |
| 13 | Cloudy | Hot | Normal | FALSE | Yes |
| 14 | Rainy | Mild | High | TRUE | No |

$$Entropy(S) = \sum_{i=1}^{n} - pi * \log_2 pi$$

$$Gain(S, A) = Entropy(S) - \sum_{i=1}^{n} \frac{|S_i|}{|S|} * Entropy(Si)$$




Entropy (Total) = Entropy No Total + Entropy Yes Total

= (-4/14 * log2(4/14) + (- 10/14 * log2(10/14))

= 0,516387121 + 0,346733448

= 0,863120569

Gain (Outlook) = Entropy Total – (Proporsi Entropy Cloudy + Proporsi Entropy Rainy + Proporsi Entropy Sunny)

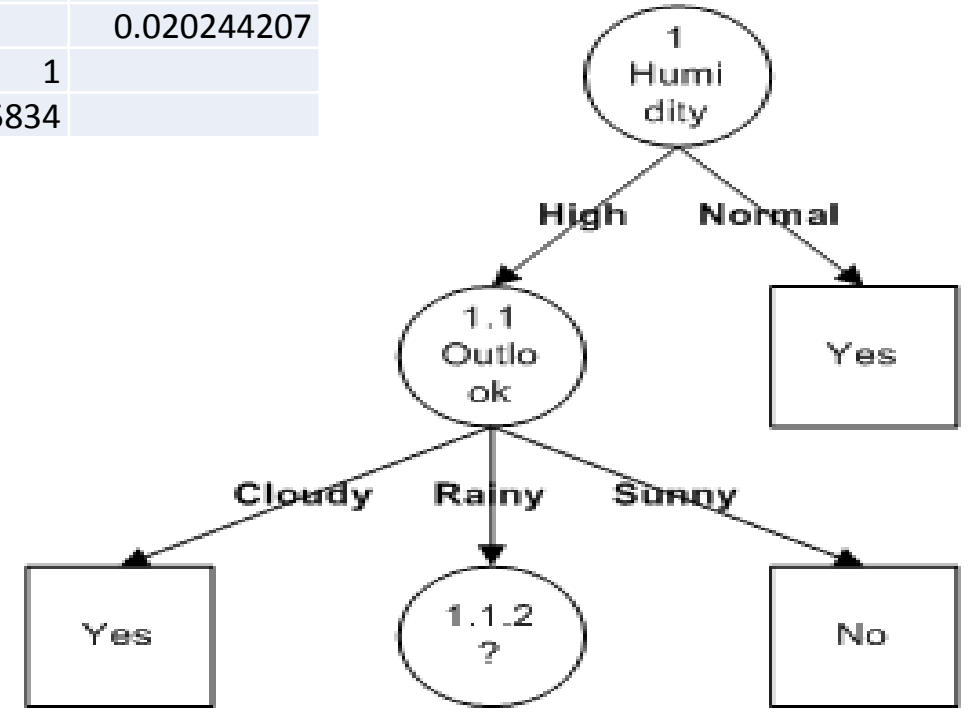
= 0,863120569 – (4/14 * 0 + 5/14 * 0,721928095 + 5/14 * 0,980950594)

= 0,863120569 – ( 0 + 0,257831462 + 0,346768069)

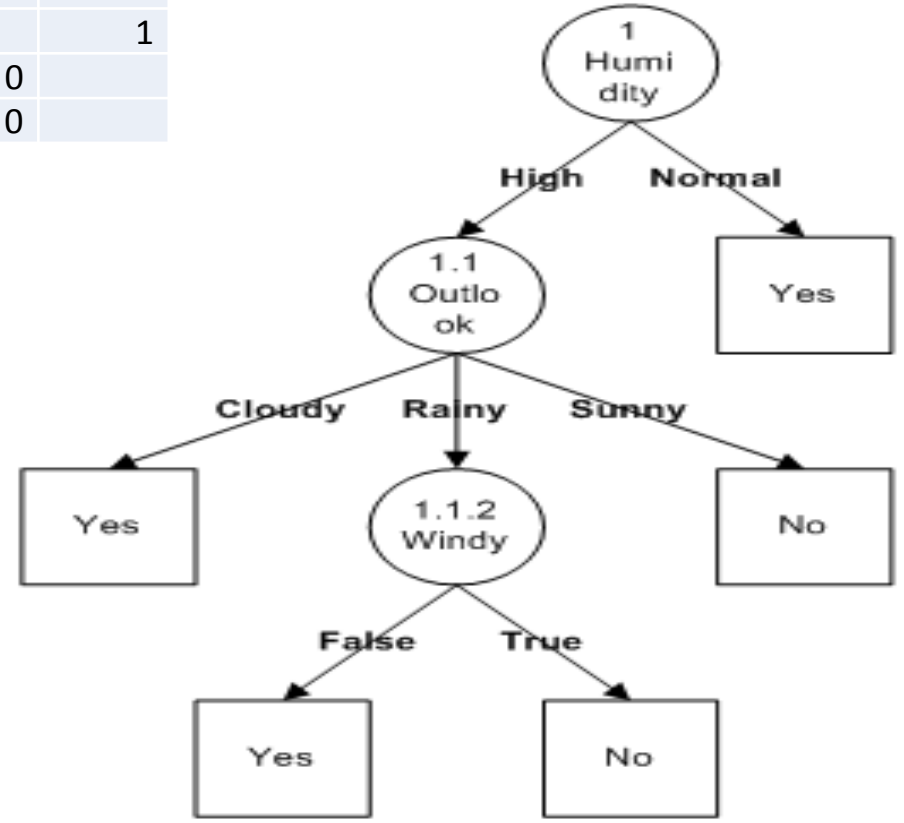
= 0,25852104

| Node | | | Jml Kasus (S) | No ($S_1$) | Yes ($S_2$) | Entropy No | Entropy Yes | Entropy | Pengurang Entropy | Gain |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | TOTAL | | 14 | 4 | 10 | 0,516387121 | 0,346733448 | 0,863120569 | | |
| | OUTLOOK | | | | | | | | | 0,25852104 |
| | | CLOUDY | 4 | 0 | 4 | #NUM! | 0 | | 0 | |
| | | RAINY | 5 | 1 | 4 | 0,464385619 | 0,257542476 | 0,721928095 | 0,257831462 | |
| | | SUNNY | 5 | 3 | 2 | 0,442179356 | 0,528771238 | 0,970950594 | 0,346768069 | |
| | TEMPERATURE | | | | | | | | | 0,18385093 |
| | | COOL | 4 | 0 | 4 | #NUM! | 0 | | 0 | |
| | | HOT | 4 | 2 | 2 | 0,5 | 0,5 | 1 | 0,285714286 | |
| | | MILD | 6 | 2 | 4 | 0,528320834 | 0,389975 | 0,918295834 | 0,393555357 | |
| | HUMIDITY | | | | | | | | | 0,3705065 |
| | | HIGH | 7 | 4 | 3 | 0,46134567 | 0,523882466 | 0,985228136 | 0,492614068 | |
| | | NORMAL | 7 | 0 | 7 | #NUM! | 0 | | 0 | |
| | WINDY | | | | | | | | | 0,00597771 |
| | | FALSE | 8 | 2 | 6 | 0,5 | 0,311278124 | 0,811278124 | 0,4635875 | |
| | | TRUE | 6 | 4 | 2 | 0,389975 | 0,528320834 | 0,918295834 | 0,393555357 | |

| Node | | | Jml Kasus (S) | No (S1) | Yes (S2) | Entropy | Gain |
|---|---|---|---|---|---|---|---|
| 1.1 | HUMIDITY-HIGH | | 7 | 4 | 3 | 0.985228136 | |
| | OUTLOOK | | | | | | 0.69951385 |
| | | CLOUDY | 2 | 0 | 2 | 0 | |
| | | RAINY | 2 | 1 | 1 | 1 | |
| | | SUNNY | 3 | 3 | 0 | 0 | |
| | TEMPERATURE | | | | | | 0.020244207 |
| | | COOL | 0 | 0 | 0 | 0 | |
| | | HOT | 3 | 2 | 1 | 0.918295834 | |
| | | MILD | 4 | 2 | 2 | 1 | |
| | WINDY | | | | | | 0.020244207 |
| | | FALSE | 4 | 2 | 2 | 1 | |
| | | TRUE | 3 | 2 | 1 | 0.918295834 | |



| Node | | | Jml Kasus (S) | No (S1) | Yes (S2) | Entropy | Gain |
|---|---|---|---|---|---|---|---|
| 1.1.2 | HUMIDITY-HIGH dan OUTLOOK -RAINY | | 2 | 1 | 1 | 1 | |
| | TEMPERATURE | | | | | | 0 |
| | | COOL | 0 | 0 | 0 | 0 | |
| | | HOT | 0 | 0 | 0 | 0 | |
| | | MILD | 2 | 1 | 1 | 1 | |
| | WINDY | | | | | | 1 |
| | | FALSE | 1 | 0 | 1 | 0 | |
| | | TRUE | 1 | 1 | 0 | 0 | |



# LATIHAN

# Buat pohon keputusan dengan Algoritma C4.5 dengan goal variabel adalah Jumlah

| No | Libur | Cuaca | Minggu_Ke | Jumlah |
|---|---|---|---|---|
| 1 | Ya | Cerah | 1 | Banyak |
| 2 | Ya | Hujan | 1 | Sedikit |
| 3 | Ya | Hujan | 1 | Sedikit |
| 4 | Tidak | Cerah | 1 | Banyak |
| 5 | Tidak | Cerah | 2 | Banyak |
| 6 | Tidak | Hujan | 2 | Sedikit |
| 7 | Tidak | Cerah | 2 | Sedikit |
| 8 | Tidak | Hujan | 3 | Sedikit |

**Diket:**

| | | | |
|---|---|---|---|
| **$\log_2$ 0/1:** | **tak terhingga** | **log2 2/5:** | **-1.321928095** |
| **$\log_2$ 0/4:** | tak terhingga | log2 3/4: | -0.415037499 |
| **$\log_2$ 1/1:** | 0 | log2 3/5: | -0.736965594 |
| **$\log_2$ 1/3:** | -1.584962501 | log2 3/8: | -1.415037499 |
| **$\log_2$ 1/4:** | -2 | log2 4/4: | 0 |
| **$\log_2$ 2/3:** | -0.584962501 | log2 5/8: | -0.678071905 |
| **log2 2/4:** | -1 | log2 5/8: | -0.678071905 |